

# Dasari Naga Raju

Andhra Pradesh, India | | WSI Biomarker Tool

## Research Interests

Computational pathology researcher with deep expertise in developing scalable AI and ML workflows for biomarker discovery, quantitative tissue phenotyping, and precision medicine from H and E whole slide images. Experienced in weakly supervised MIL architectures, spatial priors, and end to end computational imaging for molecular biomarker prediction and clinical outcome modelling. Proven track record in building reproducible image analysis pipelines integrating pathology foundation models with spatial and genomic features for cross site generalization across diverse tumor cohorts.

## Education

### IIIT Bangalore

M.S. by Research, Information Technology | *Research Area: Computational Pathology*  
*Degree structure: 59 credits supervised thesis research | 16 credits coursework (75 total)*

Sept 2021 – July 2024

**CGPA: 3.16 / 4.0**

### Jawaharlal Nehru Technological University Gurajada, Vizianagaram

B.Tech., Electronics and Communication Engineering

July 2016 – Sept 2020

**CGPA: 7.6 / 10**

## M.S. Thesis

### DeepNucleiNet — Utilizing Domain Adaptation for Deep Learning Based Necrosis Detection in Meningioma and Glioblastoma

[\[Abstract\]](#) [\[Explainability Figures\]](#)

*Supervised by Prof. T.K. Srikanth, EHRC, IIIT Bangalore · Lead Technologist, Tele MANAS, Govt. of India*

- Designed a domain-adaptive CNN exploiting cell-level phenotyping features — nuclei morphology, density, and spatial sparsity — as histology-invariant tissue descriptors for necrosis segmentation and quantification, motivated by NIMHANS clinical observations where staining variability caused complete model failure on standard architectures.
- Demonstrated cross-tumor generalization of the computational imaging pipeline trained exclusively on NIMHANS Meningioma WSIs: **97% accuracy** on TCGA Glioblastoma, **100% recall** on Low-Grade Glioma with **zero target-domain supervision**, correctly identifying LGG as necrosis-absent in line with WHO tumor grading criteria.
- Developed a reproducible, region-wise WSI analysis workflow enabling pathologists to localize, quantify, and report necrotic tissue burden across regions ranging from **997 to 97,850  $\mu\text{m}^2$** ; clinically validated and deployed as a decision support tool at **NIMHANS Hospital, Bangalore**.
- Grad-CAM revealed standard CNNs attended to uninformative tissue regions; introducing cell-level spatial features visibly redirected model attention toward biologically relevant necrotic structures, demonstrating that domain knowledge improves both quantitative accuracy and interpretability.
- t-SNE analysis confirmed progressive class separability from baseline through XceptionNet to DeepNucleiNet; validated robustness across five grid initialization offsets with consistent recall 0.95–0.97, achieving **45% F1** and **74.7% accuracy** improvement over baseline.

## Accepted Publications

1. **Dasari Naga Raju**, “Conserved Immune Topology Improves Pathology Foundation Model Generalization for Cross-Cancer MSI-H Prediction,” *ECCV 2026 Workshop on Medical Foundation Models and Benchmarks (MedFMB)*.
2. **Dasari Naga Raju**, “What Do Pathology Foundation Models Learn Beyond Gleason Grade in Prostate Cancer?” *MICCAI 2026 Workshop on Computational Pathology with Multimodal Data (COMPAYL)*.
3. **Dasari Naga Raju**, Ramesh Kestur, T.K. Srikanth, Shilpa Rao, “DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology,” *MICCAI 2026 Workshop on BrainWorks*.
4. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, “Self-Supervised Denoising Reconstruction Outperforms Foundation Models for Capsule Endoscopy Abnormality Classification,” *MICCAI 2026 Workshop on AI for Safe Surgery (SafeSurg)*.
5. **Dasari Naga Raju**, V.K. Kancharla, P. Kaveti, “Surgical Foundation Model Ensembles for Predicting Annotator Disagreement in Critical View of Safety Assessments,” *CREATE: Clinical-driven Robotics and Embodied AI TEchnology*.
6. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, “Weave Before You Weigh: Interleaved Head Attention for Multimodal Physiological Stress Recognition in the Wild,” *ECCV 2026 11th Workshop and Competition on Affective & Behavior Analysis in-the-wild*.

## Research Experience

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### Independent Computational Pathology Researcher

Jan 2025 – Present

- Proposed a grade-disentangled MIL framework using adversarial disentanglement to reduce Gleason grade dependence in learned imaging representations, combined with clinical variables via late-fusion Cox modelling; achieved superior C-index over both clinical and imaging-only baselines on TCGA-PRAD, and significantly separated BCR-free survival curves within intermediate-grade patients where grade-based stratification provides little resolution. **Accepted at MICCAI 2026 Workshop on Computational Pathology with Multimodal Data (COMPAYL).**
- Proposed Conserved Immune Topology (CIT), a lightweight spatial representation augmenting foundation-model embeddings with biologically motivated immune descriptors encoding tertiary lymphoid structures, peritumoral immune reactions, TIL density, and immune-tumour mixing without requiring annotations or target-domain data; zero-shot cross-cancer transfer improved TransMIL AUC from 0.6627 to 0.7161 across CPTAC-COAD and TCGA-STAD under scanner and organ-level distribution shift. **Accepted at ECCV 2026 Workshop on Medical Foundation Models and Benchmarks (MedFMB).**

### Project Scientist (Contract), Medical College of Wisconsin, Remote(Contract )

Feb 2025 – Present

*Project Supervisor: Dr. Pradeep Javarayee, MD MBA, Epileptologist, Children's Wisconsin & Medical College of Wisconsin*

- Developing an AI-based model to improve seizure detection across multiple hospital sites, with a focus on achieving strong cross-site generalization performance.

### Research Associate, MINRO Lab, IIIT Bangalore

Aug 2024 – Jan 2025

- Validated DeepNucleiNet across diverse WSI cohorts (Meningioma, TCGA-GBM, NIMHANS GBM, TCGA-LGG, DeepHisto, TIGER), achieving F1-scores of 0.943 on GBM, 0.843 on DeepHisto, and 0.635 on TIGER without retraining.
- Demonstrated grading-aware performance via a controlled experiment on Low-Grade Glioma, correctly producing zero necrosis predictions across 85,000+ tiles, confirming alignment with WHO grading criteria.

### Clinical Data Collaborator, NIMHANS Hospital × IIIT Bangalore

Sept 2021 – Aug 2022

- Led year-long on-site collaboration for WSI data management, gigapixel tile storage, and expert annotation of Meningioma, Glioblastoma, and Low-Grade Glioma slides under **Prof. Anita Mahadevan** and **Dr. Shilpa Rao**, Department of Neuropathology, NIMHANS Bangalore.
- Supported neuropathologist labeling of necrotic foci, cell populations (RBCs, lymphocytes), white space, and tissue regions of interest using **ImageScope**, acquiring deep domain knowledge of H&E stain interpretation and WHO tumor grading that directly informed the cell-phenotyping feature design in DeepNucleiNet.

### Research Associate, Mirai Research Innovation Institute, Japan

Oct 2020 – Aug 2021

- Developed a scalable CNN-based defect detection pipeline using ResNet50 with transfer learning for identifying missing electronic components in digitized PCB images via sliding window classification, improving accuracy from **82% to 94%**.

### Undergraduate Research Assistant, JNTU Vizianagaram

Oct 2019 – Sept 2020

- Developed a CNN-based object detection framework for nighttime vehicle classification on real-time traffic images, achieving **94% accuracy**.

## Technical Skills

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|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Digital Pathology</b> | OpenSlide, QuPath, ImageScope, HALO (familiar), StarDist (cell segmentation), MNE-Python                                |
| <b>Spatial Analysis</b>  | Biologically grounded spatial priors, tissue-level spatial feature extraction, tumour microenvironment characterisation |
| <b>Deep Learning</b>     | PyTorch, TensorFlow/Keras, Scikit-Learn, Vision Transformers, UNI2-h, Virchow2, MIL architectures                       |
| <b>Image Analysis</b>    | Tissue segmentation, cell phenotyping, object detection, representation learning, Grad-CAM, t-SNE                       |